Face Detection

1. Course Content

- Run example programs, use the mediapipe framework to detect faces.

2. Program Startup

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

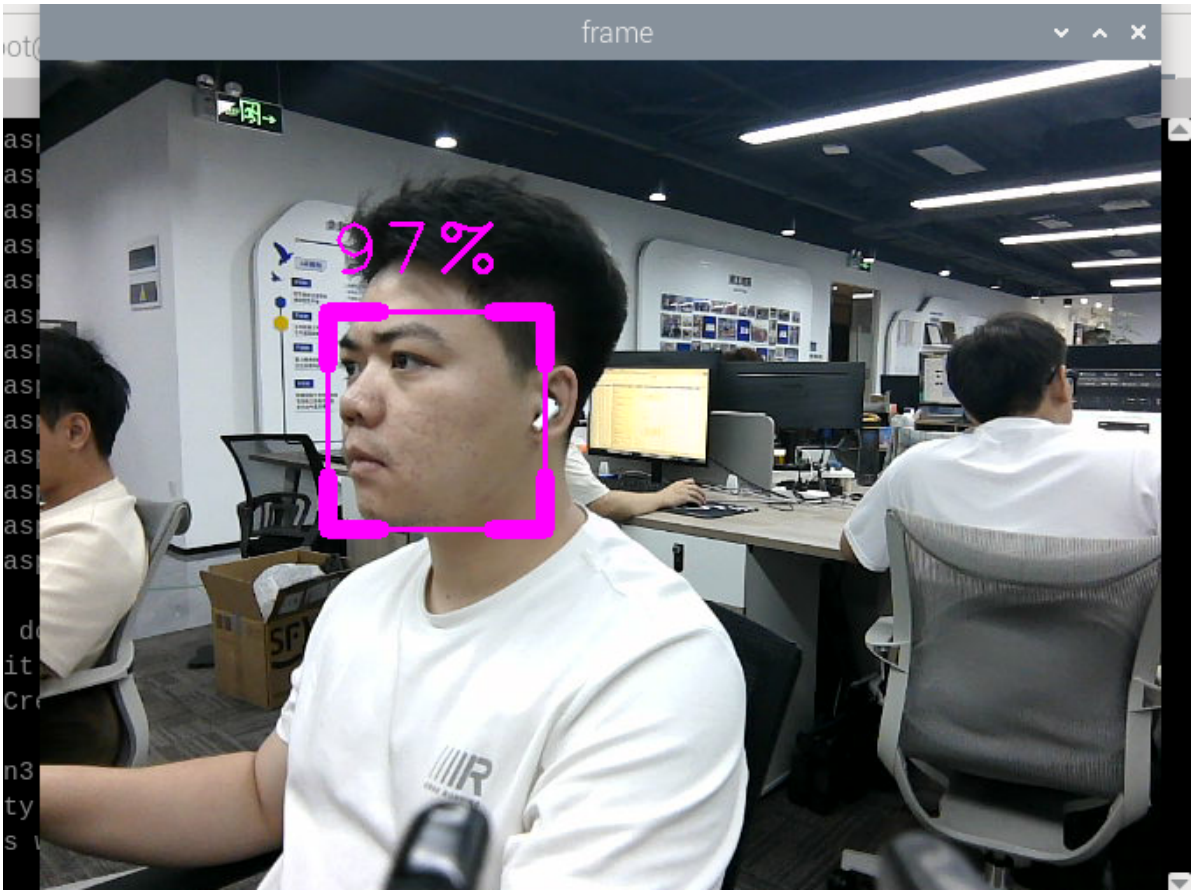
Open a terminal inside the container,

```
exec_m3
```

- Run face detection node in terminal

```
ros2 launch m3_bringup mediapipe_demo.launch.py demo:=FaceDetector
```

After the program runs, as shown in the figure below, detected faces will be boxed and the detection score will be displayed. The higher the score, the more likely it is recognized as a face.



3. Core Code Analysis

Program code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/mediapipe/05_FaceDetection.py
```

Import the required library files

```
import time
import cv2 as cv
import numpy as np
import rclpy
from rclpy.node import Node
#import mediapipe library
import mediapipe as mp
from cv_bridge import CvBridge
from sensor_msgs.msg import Image
from arm_msgs.msg import ArmJoints
import cv2
```

Initialize data and define publishers and subscribers,

```
def __init__(self, name):
    super().__init__(name)
    self.minDetectionCon=0.5
    #Use the class in the mediapipe library to define a face detection object
    self.mpFaceDetection = mp.solutions.face_detection
    self.mpDraw = mp.solutions.drawing_utils
    self.facedetection =
self.mpFaceDetection.FaceDetection(min_detection_confidence=self.minDetectionCon
)
    self.rgb_bridge = CvBridge()

    #Define the subscriber for color image topic
    self.sub_rgb =
self.create_subscription(Image,"/camera/color/image_raw",self.get_RGBImageCallBa
ck,100)
```

Color image callback function,

```
def get_RGBImageCallBack(self,msg):
    #Use CvBridge to convert color image message data to image data
    rgb_image = self.rgb_bridge.imgmsg_to_cv2(msg, "bgr8")
    #Pass the obtained image to the defined findFaces function for face detection
program
    frame,_ = self.findFaces(rgb_image)
    key = cv2.waitKey(1)
    cv.imshow('frame', frame)
```

findFaces function,

```
def findFaces(self, frame):
    #Convert the color space of the input image from BGR to RGB for subsequent
image processing
    img_RGB = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
    #Call the process function in the mediapipe library for image processing. In
init, the self.facedetection object was created and initialized
    self.results = self.facedetection.process(img_RGB)
    bboxs = []
    #Check if self.results.detections exists, i.e., whether face is recognized
    if self.results.detections:
        #Iterate through id and detection data
        for id, detection in enumerate(self.results.detections):
            #Get bounding box coordinates
            bboxC = detection.location_data.relative_bounding_box
            ih, iw, ic = frame.shape
            bbox = int(bboxC.xmin * iw), int(bboxC.ymin * ih), \
            int(bboxC.width * iw), int(bboxC.height * ih)
            #Store detection results
            bboxs.append([id, bbox, detection.score])
            #Call fancyDraw function to draw bounding box
            frame = self.fancyDraw(frame, bbox)
            #Display face recognition score on the image
            cv.putText(frame, f'{int(detection.score[0] * 100)}%',(bbox[0],
bbox[1] - 20), cv.FONT_HERSHEY_PLAIN,3, (255, 0, 255), 2)
    return frame, bboxs
```

fancyDraw function, draws bounding box based on detection result bbox values

```
def fancyDraw(self, frame, bbox, l=30, t=10):
    x, y, w, h = bbox
    x1, y1 = x + w, y + h
    cv.rectangle(frame, (x, y),(x + w, y + h), (255, 0, 255), 2)
    # Top left x,y
    cv.line(frame, (x, y), (x + l, y), (255, 0, 255), t)
    cv.line(frame, (x, y), (x, y + l), (255, 0, 255), t)
    # Top right x1,y
    cv.line(frame, (x1, y), (x1 - l, y), (255, 0, 255), t)
    cv.line(frame, (x1, y), (x1, y + l), (255, 0, 255), t)
    # Bottom left x1,y1
    cv.line(frame, (x, y1), (x + l, y1), (255, 0, 255), t)
    cv.line(frame, (x, y1), (x, y1 - l), (255, 0, 255), t)
    # Bottom right x1,y1
    cv.line(frame, (x1, y1), (x1 - l, y1), (255, 0, 255), t)
    cv.line(frame, (x1, y1), (x1, y1 - l), (255, 0, 255), t)
    return frame
```